CSE445 · NORTH SOUTH UNIVERSITY · 2026

TenVoices: Speech Recognition from Ten Voices

Which of 50 phrases was spoken — by a voice the system has never heard?

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THE TASK

Recognise one of 50 phrases, from unfamiliar speakers

- Input: one spoken command or sentence (“turn the lights on”).
- Output: which of the 50 phrases it was, with a confidence.
- The hard part: the test speakers are never heard in training.

RESEARCH QUESTIONS

- Which approach recognises speech most accurately?
- Does longer fine-tuning help?
- How do new speakers, accents and noise affect results?

DATA

Two published 10-speaker subsets

TENVOICES-FSC · MAIN

From Fluent Speech Commands (Lugosch et al., 2019): 10 speakers (5 F, 5 M) × 50 smart-home commands × 2 takes = **1,000 clips**.

Many commands share most of their words: “switch the lights on” vs “off”.

TENVOICES-L2 · ACCENTS

From L2-ARCTIC (Zhao et al., 2018): 10 non-native speakers — first languages Hindi, Mandarin, Spanish, Korean, Vietnamese × 50 sentences = **500 clips**.

Selected by script with fixed rules; audio never redistributed.

METHOD

Four stages, one shared pipeline

All audio: mono, 16 kHz, silence trimmed, peak-normalised, padded to 6 s.

1 · MFCC + CLASSIC ML

13 MFCCs + Δ + ΔΔ, mean and std (78 values) → k-NN, SVM, Random Forest, XGBoost.

2 · UNTRAINED BASELINE

wav2vec 2.0 with a new, untrained head: the chance level.

3 · FINE-TUNED

wav2vec 2.0 for 3 / 8 / 15 epochs, white-noise augmentation; also WavLM, HuBERT and a frozen linear probe.

4 · WHISPER

Zero-shot speech-to-text; the transcript is matched to the nearest of the 50 phrases.

EVALUATION

Fair comparison: seen vs unseen speakers

RANDOM SPLIT

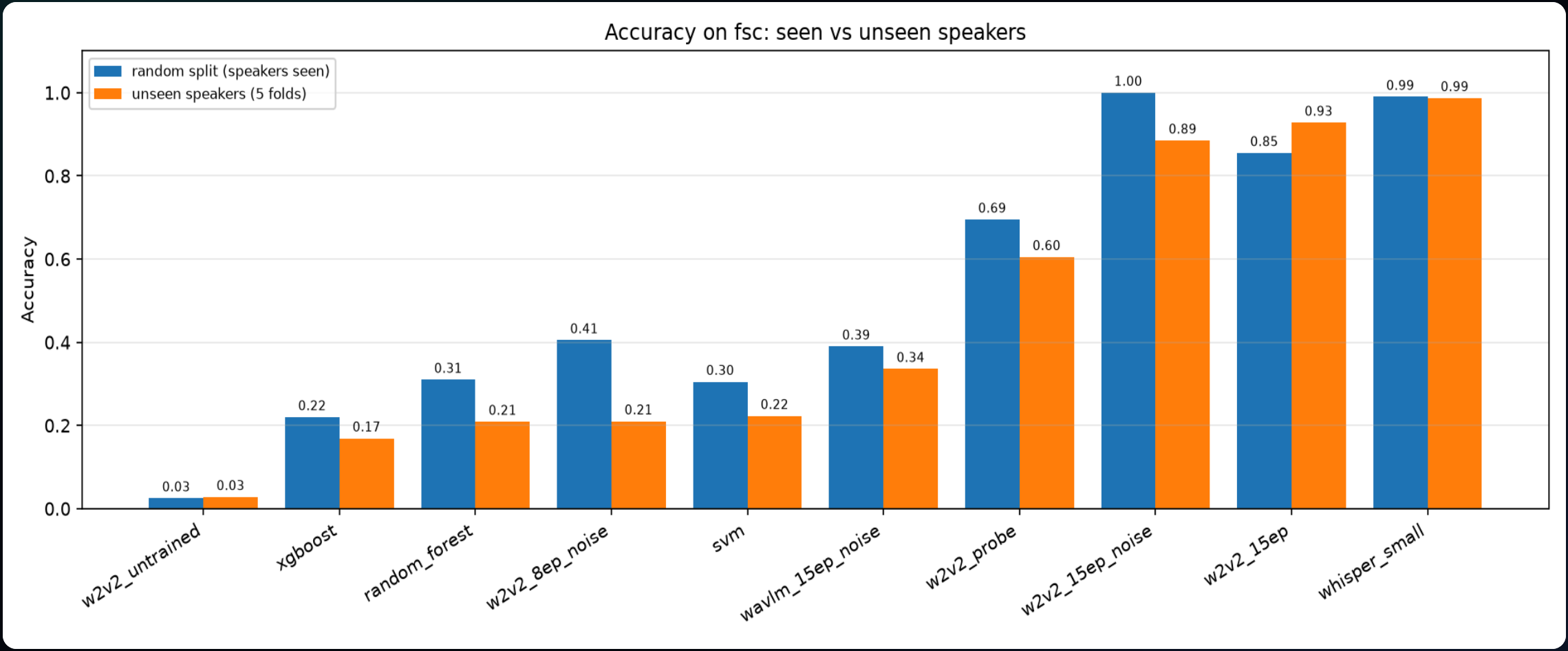
80 / 20, stratified by phrase. The test speakers were also heard in training.

SPEAKER-INDEPENDENT · HEADLINE

5 folds: 7 training, 1 validation, 2 test speakers; every speaker is tested once.

- Accuracy, macro F1, confusion matrices, accuracy per speaker; word error rate for Whisper.
- Best epoch and hyper-parameters chosen on validation data, never on the test set.
- White noise added to the test clips at 20 / 10 / 5 dB.

Accuracy on seen and unseen speakers



source: results/<run>/fsc/summary.json · figure: python -m support.visualization --results results

RESULTS

Pre-training does most of the work

The same wav2vec 2.0 Base model, on unseen speakers:

2.8%
untrained head
(chance is 2%)

→

60.5%
frozen features +
logistic regression

→

88.5%
fine-tuned
15 epochs + noise

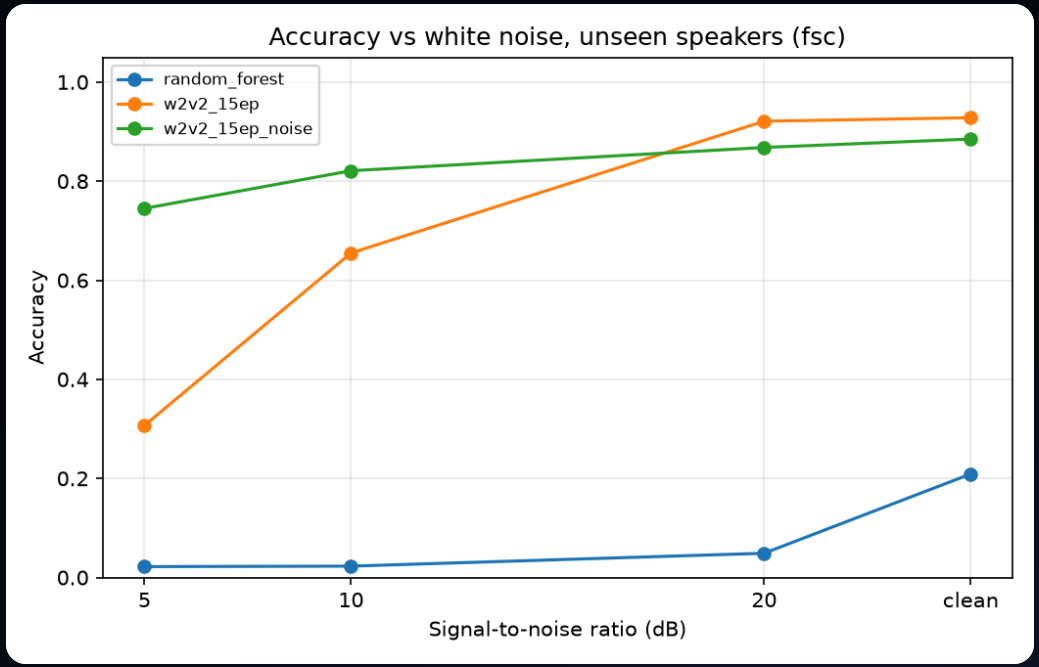
For comparison: MFCC + SVM reaches 22.3%, MFCC + Random Forest 20.9%; zero-shot Whisper 98.7%.

source: results/{w2v2_untrained,w2v2_probe,svm}/fsc · results/M1_tanvir/{w2v2_15ep_noise,whisper_small}/fsc

RESULTS · NOISE

White-noise training buys robustness

unseen speakers	clean	20 dB	10 dB	5 dB
wav2vec 2.0 + noise	88.5%	86.8%	**82.1%**	**74.5%**
wav2vec 2.0, no noise	92.8%	92.1%	65.5%	30.7%
MFCC + Random Forest	20.9%	4.9%	2.3%	2.2%



Training with noise costs a little on clean audio but keeps the model usable at 5 dB; MFCC features collapse to chance as soon as noise is added.

source: results/M1_tanvir/noise/*_fsc.csv

RESULTS · TENVOICES-L2

Accents: right sentence, more word errors

unseen first language	accuracy
MFCC + Random Forest	17.0%
Whisper small, zero-shot	**99.6%**

Whisper word error rate	unseen folds
TenVoices-FSC (native)	3.9%
TenVoices-L2 (non-native)	10.0%

Whisper still picks the right sentence almost every time, but its transcripts contain more errors for non-native speakers (hypothesis H5, tested formally in the report).

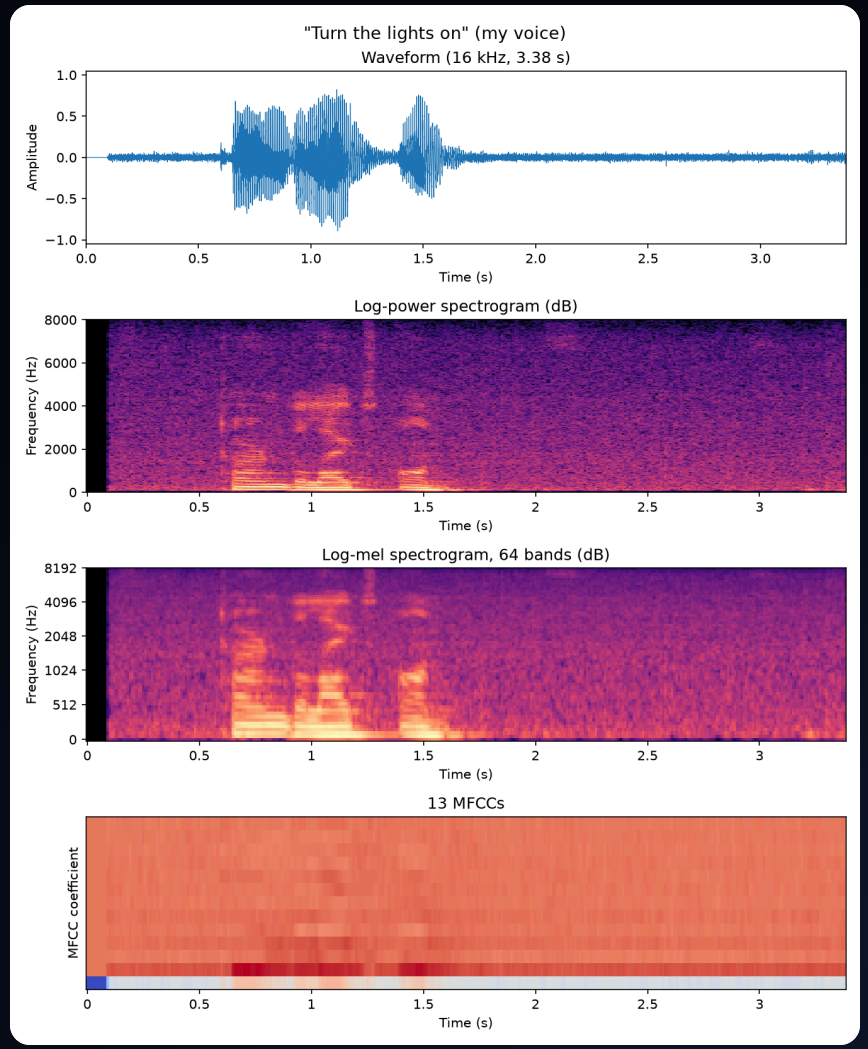
source: results/M1_tanvir/{random_forest,whisper_small}/{fsc,l2arctic}/summary.json

DEMO

Live demo: speak, see, recognise

- python app.py: record or upload a phrase.
- Shows what the model receives: waveform, spectrograms, MFCCs.
- Top-5 phrases with confidence; optional Whisper transcript.

a new voice, 3 commands	correct
MFCC + SVM	0/3
Linear probe	0/3
Whisper, zero-shot	**3/3**



source: results/demo_check/shafil_voice_fsc.csv (3 clips: an illustration, not a test)

CONCLUSIONS

What we learned

- Self-supervised pre-training makes 50 commands learnable from 7 training speakers.
- MFCC models depend heavily on the speaker and fail in noise.
- Noise augmentation trades a little clean accuracy for robustness.
- Zero-shot Whisper is the strongest and most portable, but accents raise its word errors.

FUTURE WORK

- Fine-tune Whisper or wav2vec 2.0 (CTC) on non-native speech.
- Multilingual pre-training (XLS-R).
- Bangladeshi speakers reading the same commands.
- All 97 FSC speakers: how accuracy scales with speakers.

THANK YOU

Questions?

github.com/minhaz-42/Voices-Speech-Recognition-445